Introduction to Machine Learning

Apr 21, 2025 – Machine Learning

Prashant Poudel

Resources

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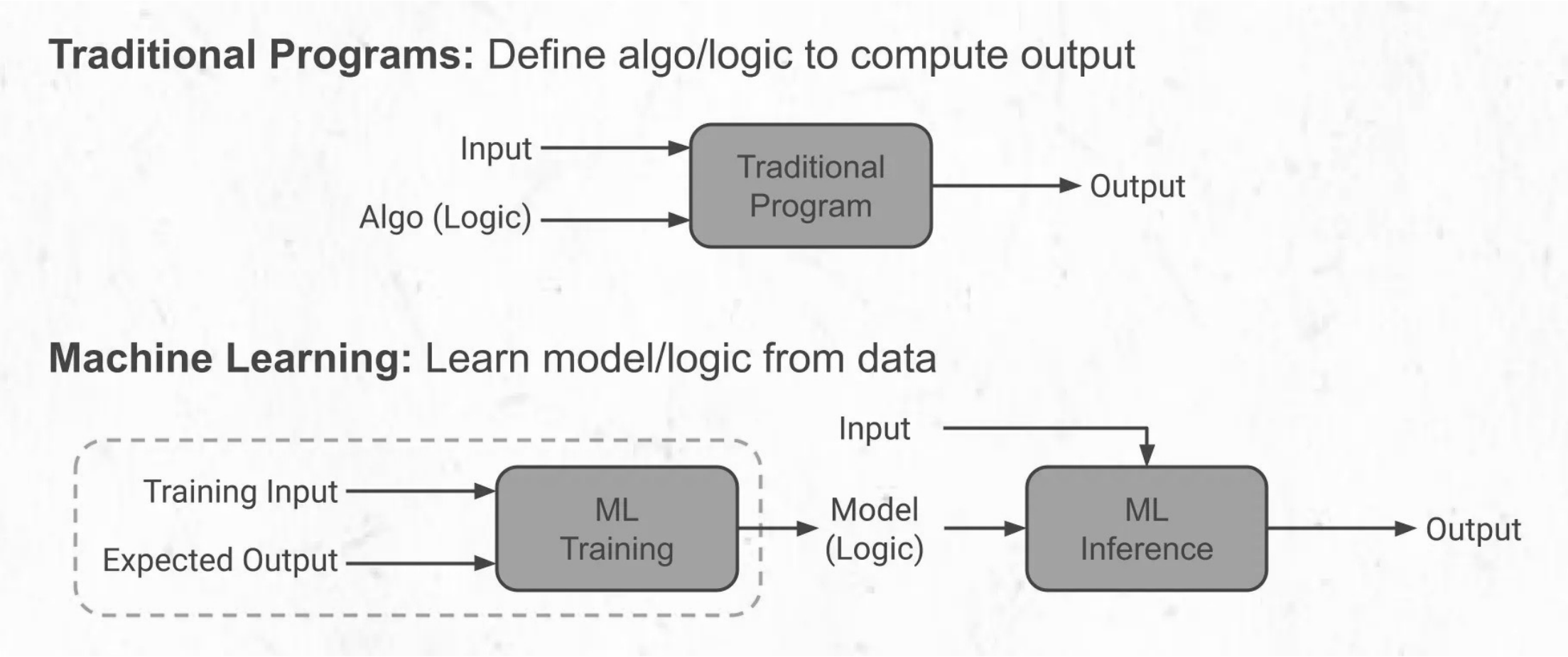
Machine Learning

- Machine Learning (ML) involves teaching a computer to recognize patterns by example, rather than programming it with specific rules. These patterns can be found within data.
- In other words, ML is about creating algorithms (or a set of rules) that learn complex functions (or patterns) from data and make predictions on it .
- These systems can be reused example to recognize other objects with new data. Same code.
- ML is all about predicting things essentially. It is intelligent because:
  1. It takes some data (to train the system)
  2. Learns patterns from this data
  3. Classifies new data it has not seen before for a best guess of what it probably is, based on knowledge gained from step 2.

Machine Learning



ML
History

1805

Linear Regression

(The method of least squares): Developed by Adrien-Marie Legendre (1805) and Carl Friedrich Gauss (1809).

1957

Perceptron

(Precursor to neural networks): Invented by Frank Rosenblatt.Simple model simulating a neuron for binary classification tasks.

1959

Reinforcement Learning

Inventor: Richard Bellman. Algorithmic method teaching agents to make decisions through rewards and penalties in environments.

1964

Support Vector Machines

(SVMs): Introduced by Vladimir Vapnik and Alexey Chervonenkis. excel in handwriting recognition, face detection, and various classification tasks.

1986

Backpropagation for Neural Networks

Popularized by David Rumelhart, Geoffrey Hinton, and Ronald Williams.Essential algorithm for training neural networks through gradient descent optimization.

1990

Boosting Algorithms

(AdaBoost) Introduced by Yoav Freund and Robert Schapire. Improves prediction by combining multiple weak learning models.

1995

Random Forests

Robust classification and regression through decision tree-like model ensembles. Introduced by Tin Kam Ho's

1997

RNN and LSTM

Inventors: Various, notable contributions by Sepp Hochreiter, Jürgen Schmidhuber (LSTM). Excelled in sequential data processing, enhancing NLP and speech

2012

Deep Convolutional Neural Networks

Inventors: Alex Krizhevsky, Ilya Sutskever, Geoffrey Hinton. Revolutionized image recognition with advanced, layered neural architectures.

2014

Generative Adversarial Networks

Inventor: Ian Goodfellow and colleagues. GANs have been groundbreaking in generating realistic images and videos, contributing significantly to both creative and analytical applications

2017

Transformer Networks

Inventors: Ashish Vaswani and colleagues. Pioneered NLP advancements with efficient context-aware processing.cornerstone for modern NLP tasks, leading to models like BERT, GPT series, and T5.

Evolution of Machine Learning

Machine Learning (ML) has evolved over decades, influenced by advancements in computer science, mathematics, and the availability of data. Here's a brief timeline of its evolution:

1. Rule-Based Systems (1950s – 1980s)

In the early days, systems relied on **explicitly coded rules** written by humans.

- Inspired by early AI research
- No learning from data
- Used in expert systems (e.g., MYCIN for medical diagnosis)
- *Limitation:* Could not adapt to new situations without rewriting rules.

Evolution of Machine Learning

2. Traditional Machine Learning (1990s – 2000s)

Algorithms began to learn patterns from data rather than relying on hard-coded rules.

- Emergence of **supervised**, **unsupervised**, and **reinforcement learning**
- Common algorithms: *Decision Trees, SVMs, k-NN, Naïve Bayes*
- Used in spam detection, fraud detection, and recommendation systems
- *Milestone:* Rise of statistical learning and better model generalization.

Evolution of Machine Learning

3. Big Data Era (2010s)

With more data and better computing power, ML started thriving.

- Distributed computing frameworks (like Hadoop and Spark)
- Rise of **deep learning** and **neural networks**
- Applications: speech recognition, image classification, language translation
- *Milestone:* Models like **AlexNet** (2012) revolutionized image recognition.

Evolution of Machine Learning

4. Modern ML and Deep Learning (2015 – Present)

- Focus on **deep learning**, **transformers**, and **self-supervised learning**
- Use of GPUs/TPUs for fast computation
- Applications in autonomous vehicles, healthcare, and large language models (like ChatGPT)
- *Milestone:* Introduction of **transformers** (e.g., BERT, GPT) reshaped NLP.

Evolution of Machine Learning

5. Future Trends

- **Explainable AI** and ethical ML
- **TinyML** for edge devices
- Integration with IoT, robotics, and AR/VR
- Focus on **sustainable AI** (energy-efficient models)

Machine Learning

Supervised Learning

- Labeled data
- Guessing the correct answers

Regression

Making predictions from continuous set of values

Classification

Making predictions based on finite data set

Unsupervised Learning

- Data is not labeled

Clustering

Discovering different patterns based on data set

Reinforced Learning

- Data in the form of rewards and penalties

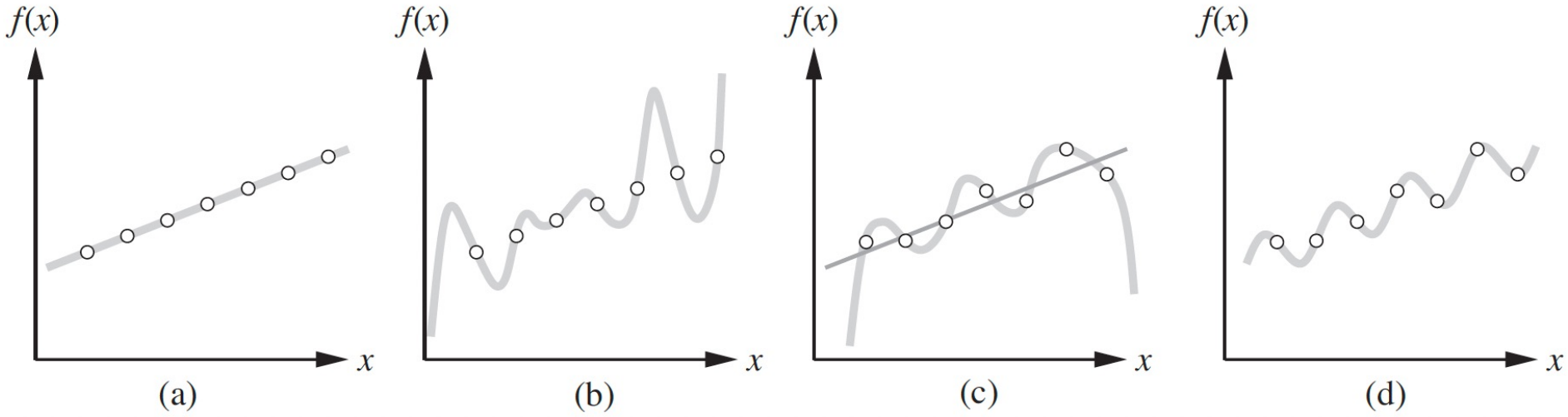
Supervised Learning

- Supervised Learning algorithms build a mathematical model of a set of data (training data) that contains both example inputs and their desired outputs.
- The goal is to learn a general rule that maps inputs to outputs through iterative optimization of an objective function.
- Supervised learning algorithms learn a function that can be used to predict the output associated with new inputs.
- Examples: Classification and regression algorithms

Supervised Learning

The task of supervised learning is this:

- Given a training set of N example input–output pairs
- $(x_1, y_1), (x_2, y_2), \ldots (x_N, y_N)$,
- where each y_j was generated by an unknown function $y = f(x)$, discover a function h that approximates the true function f. Here x and y can be any value.

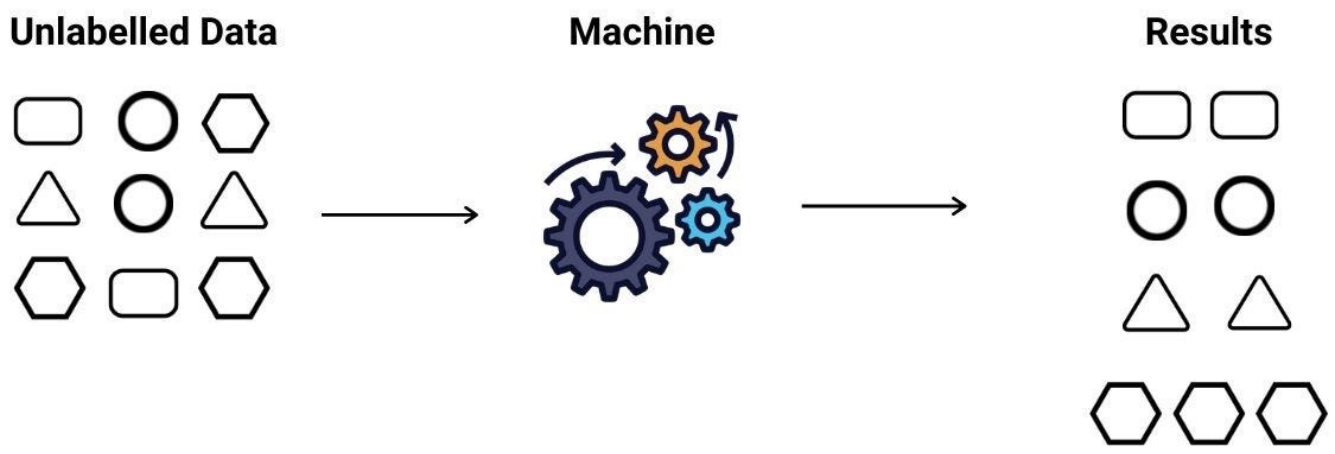


Unsupervised learning

- Unsupervised learning in artificial intelligence is a type of machine learning that doesn't need human guidance.
- Unlike supervised learning, where machines are taught with labeled data, unsupervised learning uses unlabeled data to discover patterns and insights on its own.
- The model works with raw, unlabeled data and figures out its own rules and structure by identifying similarities, differences, and patterns, all without specific instructions on how to handle each piece of data.
- Unsupervised machine learning involves three main types of tasks: clustering, association, and dimensionality reduction.

Unsupervised learning

- These algorithms are well-suited for more complex tasks, like organizing large datasets into clusters.
- They excel at finding hidden patterns in data and can help pinpoint features useful for categorizing information.
- Consider a scenario with a vast weather dataset. An unsupervised learning algorithm examines the data, recognizing patterns in data points.



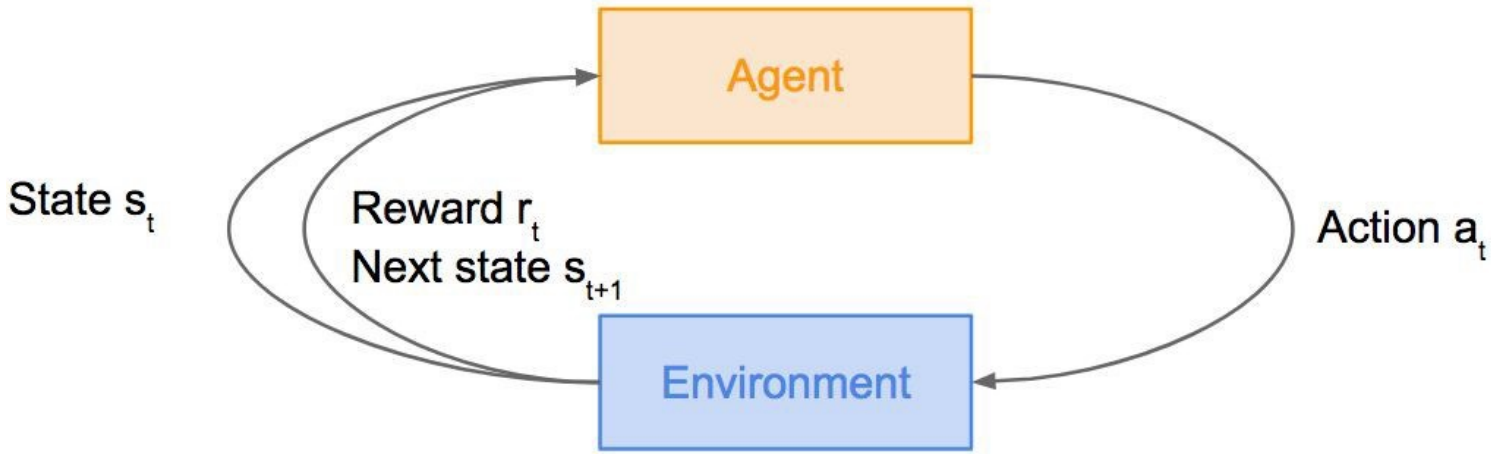
Reinforcement Learning

- Reinforcement Learning(RL) is a type of machine learning technique that enables an agent to learn in an interactive environment by trial and error using feedback from its own actions and experiences.
- Supervised learning and reinforcement learning both involve mapping inputs to outputs. Unlike supervised learning, reinforcement learning uses rewards and punishments to signal positive and negative behavior.
- In unsupervised learning, the aim is to uncover patterns in data, where as reinforcement learning focuses on finding an action model that maximizes the agent's total cumulative reward.

Reinforcement Learning

Problems involving an **agent** interacting with an **environment**, which provides numeric **reward** signals

Goal: Learn how to take actions in order to maximize reward



Reinforcement Learning

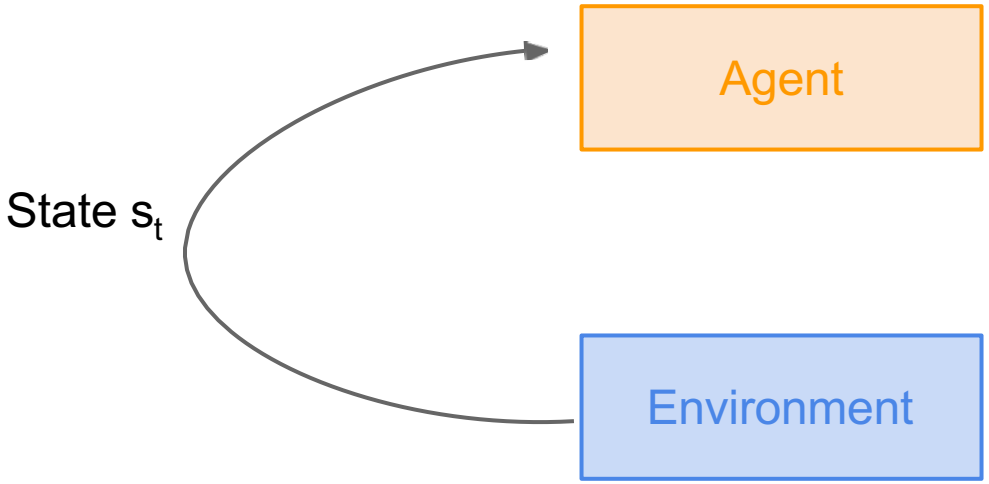
Agent: takes actions.

Agent

Environment: the world in which the agent exists and operates.

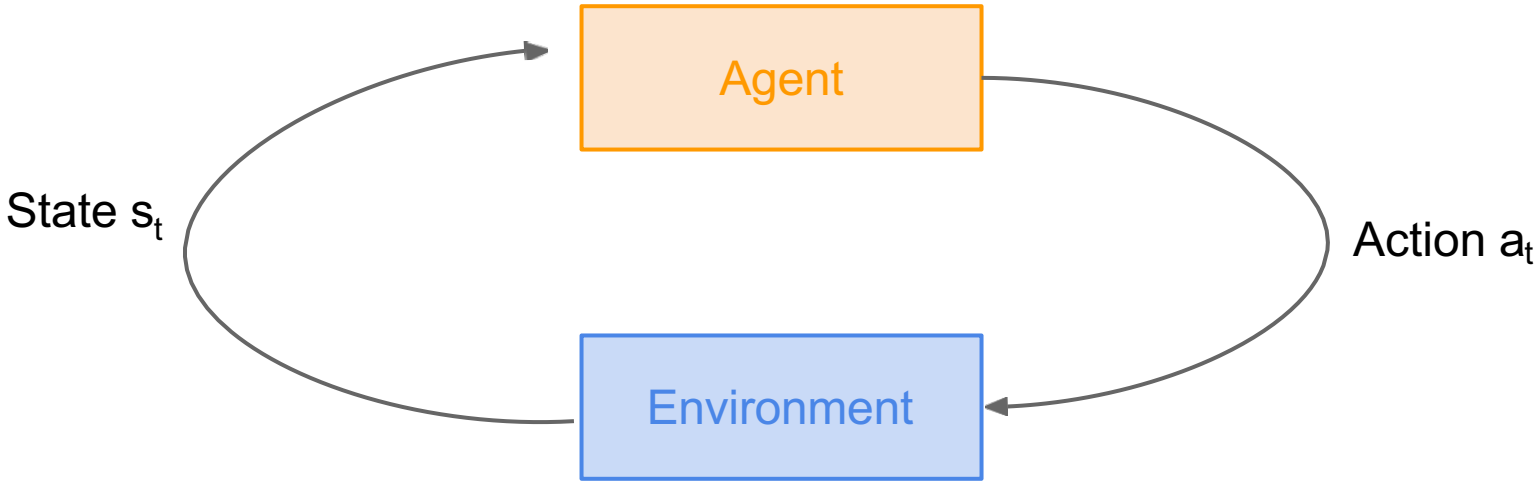
Environment

Reinforcement Learning



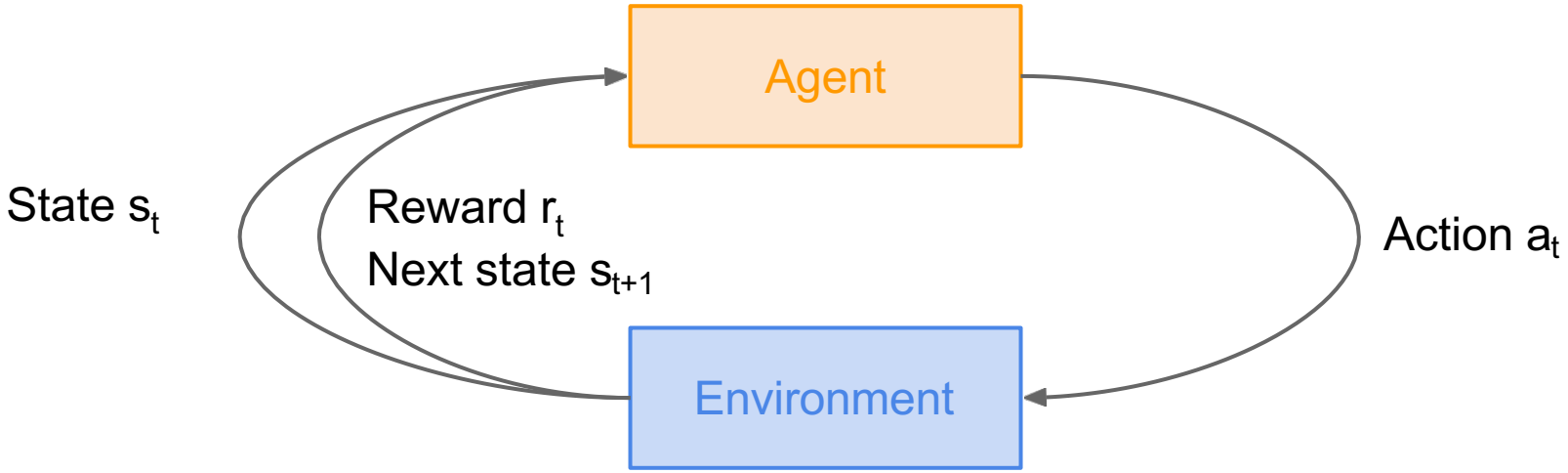
State: a situation which the agent perceives.

Reinforcement Learning



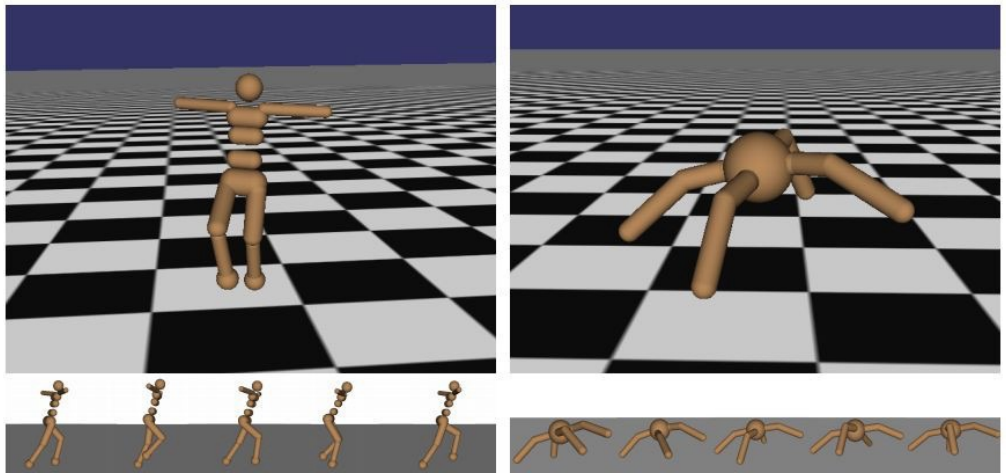
Action: a move the agent can make in the environment.

Reinforcement Learning



Reward: feedback that measures the success or failure of the agent's action.

Reinforcement Learning



Objective: Make the robot move forward

State: Angle and position of the joints
Action: Torques applied on joints
Reward: 1 at each time step upright + forward movement

Reinforcement Learning

Objective: Complete the game with the highest score

State: Raw pixel inputs of the game state
Action: Game controls e.g. Up, Down
Reward: Score increase at each time step

Reinforcement Learning: Key Elements

Policy:

- A policy dictates the behavior of the learning agent over a specific time period.
- It serves as a mapping from the perceived states of the environment to the corresponding actions the agent should take when encountering those states.
- It essentially guides the agent's decision-making process, determining its actions based on the current state of the environment.
- Policies can be deterministic or stochastic, reflecting a predetermined action or a distribution of possible actions.

Reinforcement Learning: Key Elements

Reward Function:

- The reward function plays a crucial role in defining the objectives of a reinforcement learning problem.
- It is a mathematical function that assigns a numerical score based on the state of the environment or the actions taken by the agent.
- By providing a reward signal, the reward function guides the agent towards desired outcomes.
- Agents aim to maximize cumulative rewards over time, using the reward function as a measure of success or failure.

Reinforcement Learning: Key Elements

Value Function:

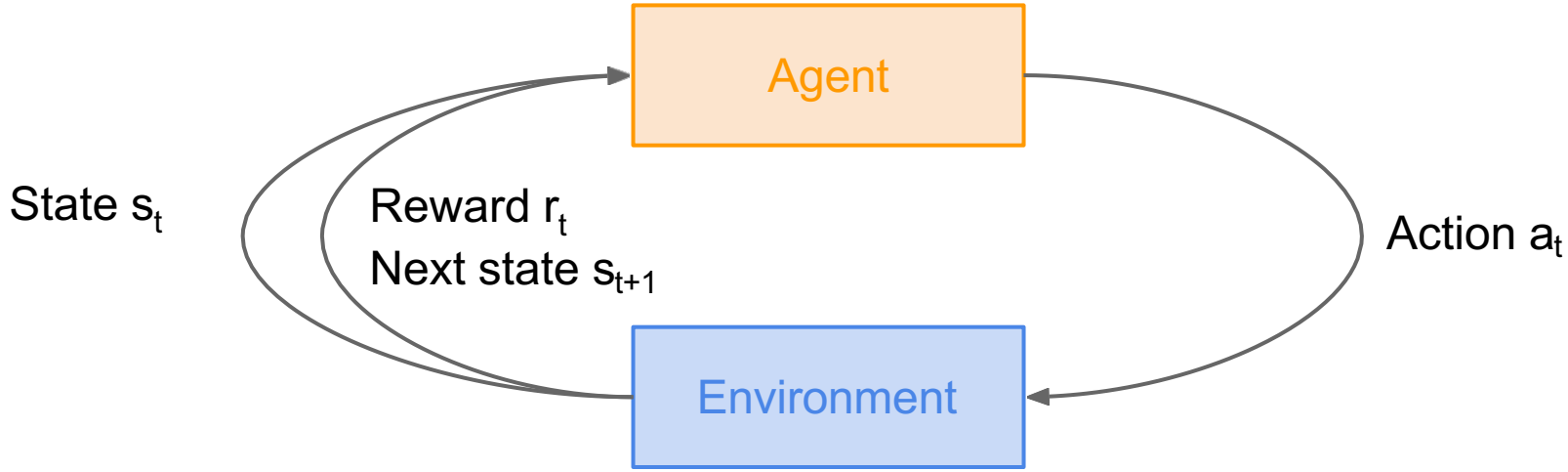
- Value functions are instrumental in determining the long-term desirability of different states.
- The value of a state is a representation of the total expected reward an agent can accrue from that state onwards.
- Value functions guide the agent in assessing the potential of states, aiding in decision-making for actions that lead to favorable long-term outcomes.
- They contribute to the agent's ability to prioritize actions that maximize cumulative rewards.

Reinforcement Learning: Key Elements

Model of the Environment:

- In reinforcement learning, models are utilized for planning and simulation purposes.
- These models attempt to capture the dynamics of the environment, predicting the possible consequences of actions taken by the agent.
- By employing a model of the environment, agents can simulate scenarios and plan actions without directly interacting with the real environment.
- This aids in learning optimal strategies and contributes to the efficiency of the learning process.

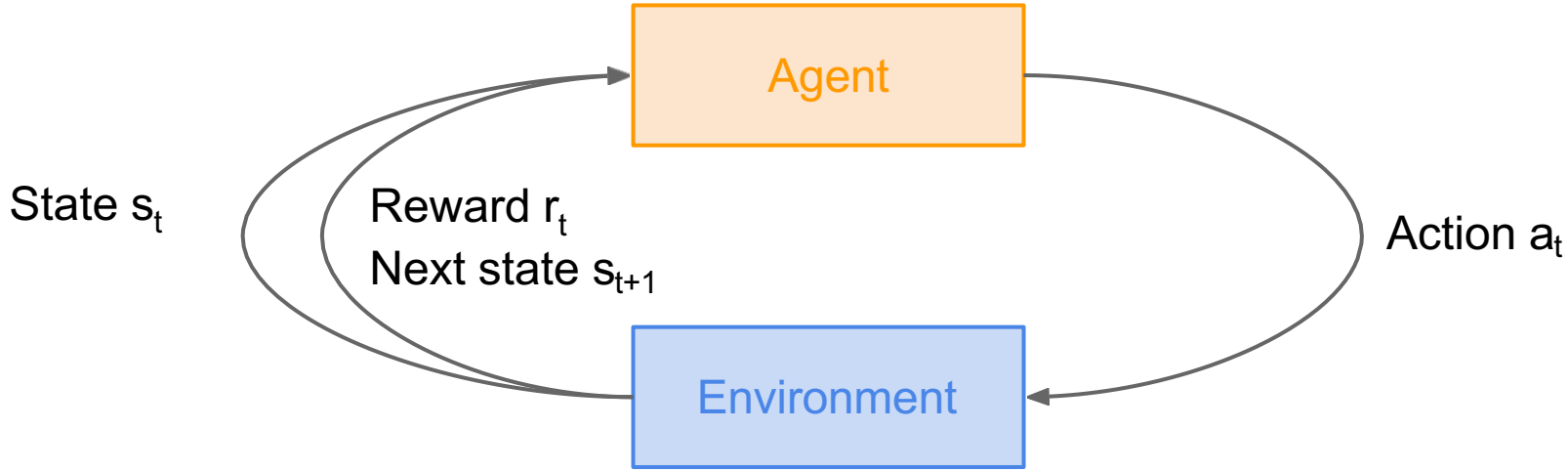
Reinforcement Learning



Total Reward

$$R_t = \sum_{i=t}^{\infty} r_i \quad \Rightarrow \quad R_t = \sum_{i=t}^{\infty} r_i = r_t + r_{t+1} \ldots + r_{t+n} + \cdots$$

Reinforcement Learning



Discounted Total Reward

$$R_t = \sum_{i=t}^{\infty} \gamma^i r_i = \gamma^t r_t + \gamma^{t+1} r_{t+1} \ldots + \gamma^{t+n} r_{t+n} + \cdots$$

γ: discount factor; $0 < \gamma < 1$

Q- Function

- The Q-function captures the expected total future reward an agent in state, **s**, can receive by executing a certain action, **a.**
- The agent learns a Q-value for each action in each state, representing how good that action is in that state.
- It uses a formula to update Q-values based on rewards and the expected future rewards.

$$\underbrace{Q^{\pi}(s_t, a_t)}_{\text{Q value for that state given that action}} = \underline{E}[\underbrace{R_{t+1} + \gamma R_{t+2} + \gamma^2 R_{t+3} + \ldots}_{\text{Expected discounted cumulative reward}} | \underbrace{s_t, a_t}_{\text{Given that state and that action}}]$$

Applications of RL

- **Game Playing:** RL is employed to train agents in playing complex games, achieving superhuman performance in titles like chess, Go, and video games.
- **Robotics:** RL enables robots to learn tasks by trial and error, adapting to different environments and optimizing their actions for specific objectives.
- **Autonomous Vehicles:** RL is used to develop algorithms for autonomous vehicles, allowing them to learn optimal navigation strategies and adapt to changing traffic conditions.
- **Finance:** RL is applied in algorithmic trading, portfolio optimization, and risk management, where agents learn to make financial decisions based on market dynamics.

Applications of RL

- **Energy Management:** RL is utilized to optimize energy consumption in smart grids, buildings, and industrial processes, leading to more sustainable and cost-effective energy use.
- **Education:** RL is employed in adaptive learning systems, where agents assist in creating personalized learning paths for students based on their performance and preferences.
- **Internet of Things (IoT):** RL is applied to optimize the control and management of IoT devices, allowing them to adapt to changing conditions and user preferences.

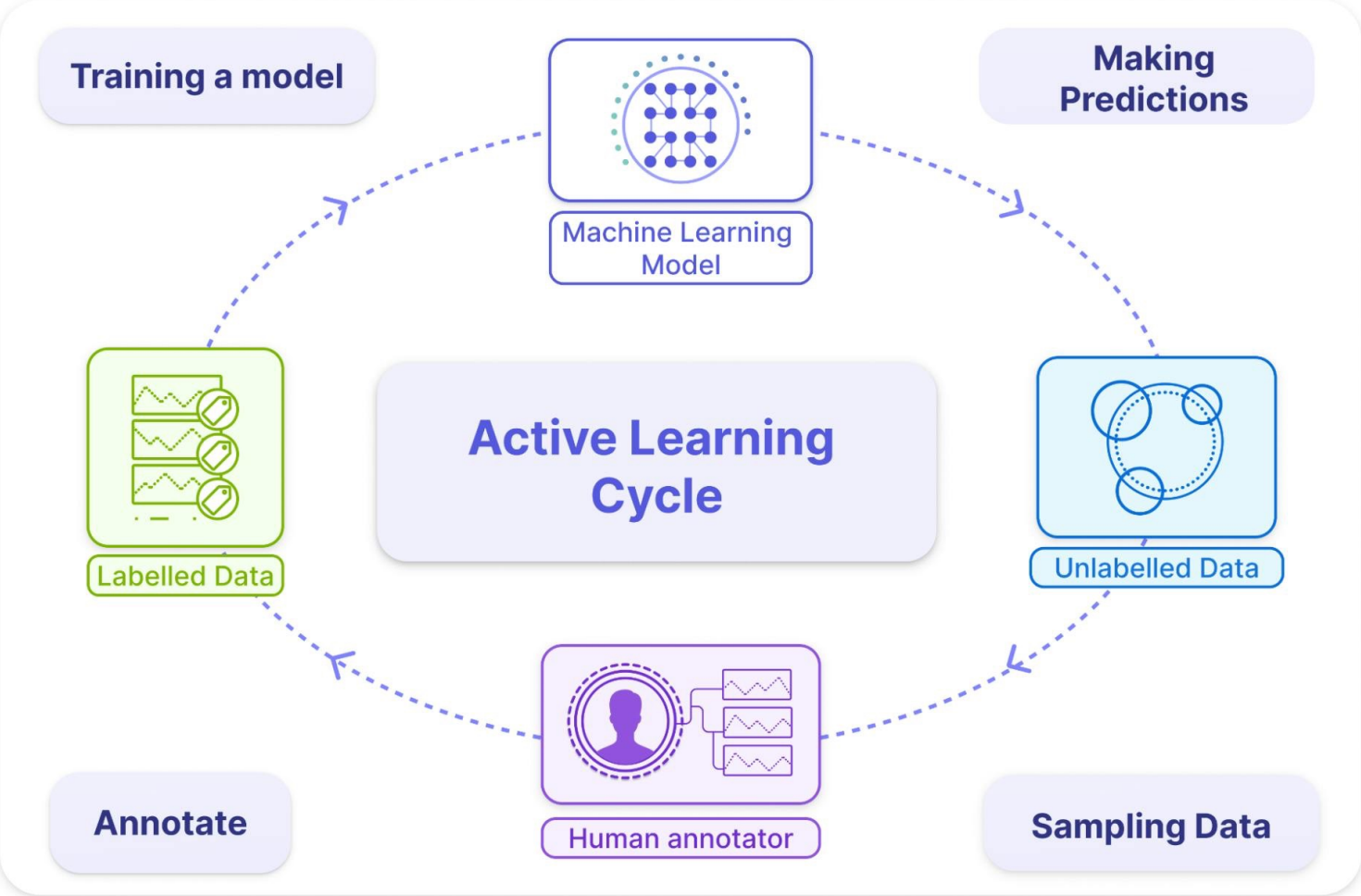
Active Learning

- Active Learning in Machine Learning is a technique where the model selectively queries the most informative data points to be labeled, instead of using all available unlabeled data.
- This helps in building accurate models with fewer labeled examples, saving time and cost.
- Why Use Active Learning?
  - Labeling data is expensive and time-consuming.
  - Most data may be redundant or easy to classify.
  - Model improves faster if it learns from uncertain or boundary cases.

How Active Learning Works

1. Start with a small labeled dataset.
2. Train an initial model.
3. Use the model to predict labels on the unlabeled dataset.
4. Select the most uncertain or informative samples.
5. Get those samples labeled by an oracle (e.g., human expert).
6. Retrain the model with the newly labeled data.
7. Repeat the process.

How Active Learning Works



Active Learning: Query Strategies

1. **Uncertainty Sampling** – Select samples where the model is least confident.
2. **Query by Committee** – Use multiple models and pick samples with the most disagreement.
3. **Expected Model Change** – Pick data that would most change the model if added.
4. **Diversity Sampling** – Ensure selected data is diverse, not just uncertain.

Active Learning: Use Cases

- Medical diagnosis (labeling images by radiologists).
- Text classification (sentiment, spam detection).
- Image recognition.
- Any domain with expensive labels.

Machine Learning Workflow

A typical ML workflow involves several key stages. These stages are often **iterative**—you may revisit earlier steps as you gather new insights or improve performance.

1. Problem Definition

Start by clearly defining the problem you want to solve and determining if ML is the right approach.

Questions to consider:

- What exactly are you trying to predict or classify?
- Do you have enough relevant data?
- What would a successful model look like in practice?

- *Example:* Predict house prices based on location, size, and features.
- *Tip:* Not all problems need ML. Rule-based systems may work better for simple, deterministic tasks.

Machine Learning Workflow (Contd…)

2. Data Collection and Preprocessing

2.1. Data Collection

Gather relevant data from various sources such as databases, sensors, surveys, or APIs. Your dataset should contain:

- Input features (independent variables)
- Target labels (for supervised learning)

Example: A dataset of past house sales with features like area, number of rooms, and sale price.

Machine Learning Workflow (Contd…)

2. Data Collection and Preprocessing

2.2 Data Cleaning & Preprocessing

Transform raw data into a suitable format for training:

- Handle missing or duplicate values
- Encode categorical variables
- Normalize or standardize numerical values
- Remove outliers or noise

- *Example:* Convert text-based location data into numeric format, fill in missing values for square footage, normalize price values.

Machine Learning Workflow (Contd…)

3. Model Selection

Choose a machine learning algorithm that fits the nature of your problem and data.

- **Classification problems** (e.g., spam detection): Use Decision Trees, SVM, k-NN, Logistic Regression, etc.
- **Regression problems** (e.g., price prediction): Use Linear Regression, Random Forest, Gradient Boosting, etc.
- **Clustering or pattern detection**: Use K-Means, DBSCAN, etc.

- *Example:* Use Linear Regression to estimate house prices, or Decision Trees to classify loan approvals.

Machine Learning Workflow (Contd…)

4. Model Evaluation and Validation

Split your dataset into:

- **Training set**: to build the model.
- **Validation set**: to tune parameters and prevent overfitting.
- **Test set**: to evaluate final model performance on unseen data.

Use appropriate metrics:

- **Classification**: Accuracy, Precision, Recall, F1-score
- **Regression**: Mean Squared Error (MSE), Root Mean Squared Error (RMSE), R^2 Score

- *Example:* Evaluate a price prediction model using RMSE. If the error is too high, adjust the model or preprocessing.

Machine Learning Workflow (Contd…)

5. Model Deployment

After achieving satisfactory performance, integrate the model into a real-world application where it can make predictions on new data.

Deployment can be:

- **Local** (e.g., within a mobile app)
- **Server-based** (e.g., REST API)
- **Cloud-hosted** (e.g., via web services)

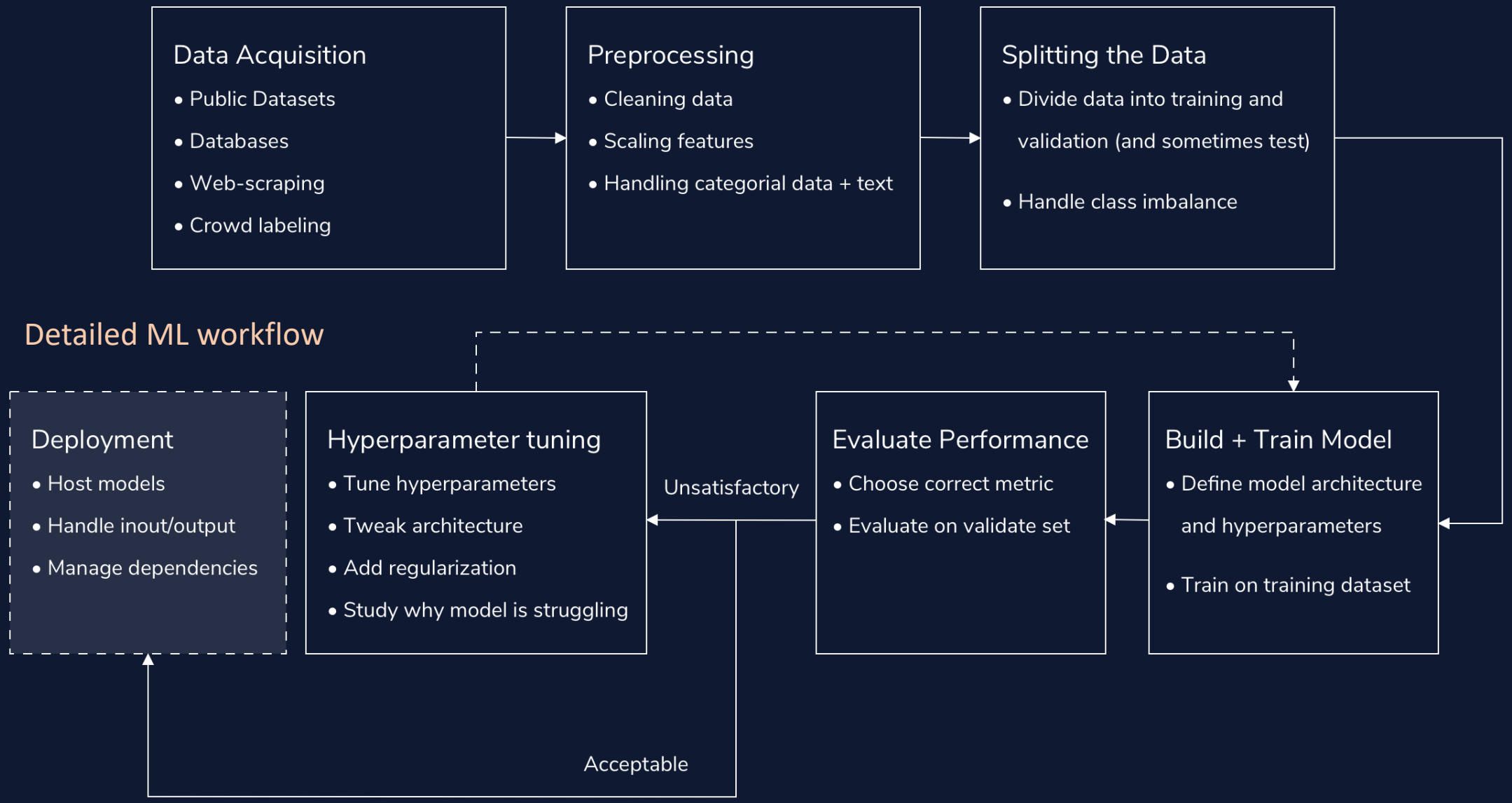
Example: Deploy a model that predicts product demand on a company dashboard or via an API that powers a recommendation system.

Machine Learning Workflow (Contd…)

6. Ongoing Monitoring and Maintenance

- Monitor predictions for accuracy and fairness.
- Retrain the model periodically with new data.
- Track performance metrics to detect model drift.
- *Example:* If user behavior changes over time, regularly retrain a recommendation model to maintain relevance.

Detailed ML workflow



Challenges in Machine Learning

Despite its power and potential, machine learning comes with several challenges that can affect the performance, reliability, and acceptance of models.

1. **Data Quality Issues:** Machine learning models heavily rely on the quality and quantity of data. Poor data can lead to inaccurate or biased results.

   Common issues:

   - Missing or incomplete data
   - Noisy or inconsistent values
   - Imbalanced datasets (e.g., too few positive examples)
   - Outdated or irrelevant features

- *Example:* A medical diagnosis model trained on outdated patient records may perform poorly on current cases.
- *Solution:* Regular data audits, cleaning pipelines, and data augmentation techniques.

Challenges in Machine Learning (Contd…)

2. Computational Complexity: Training complex models, especially on large datasets or using deep learning, can be computationally expensive.

Challenges include:

- Long training times
- Need for powerful hardware (GPUs/TPUs)
- High energy consumption
- Scalability for real-time inference

- *Example:* Training a deep neural network for image classification might take several hours or days on limited hardware.
- *Solution:* Use cloud resources, optimize algorithms, or consider simpler models when possible.

Challenges in Machine Learning (Contd…)

3. Interpretability and Explainability: Many powerful ML models (e.g., deep neural networks, ensemble methods) act like "black boxes," making it hard to understand how decisions are made.

Why it matters:

- Difficult to debug or improve the model
- Users may not trust the model
- Legal and regulatory compliance (e.g., in finance or healthcare)

- *Example:* A bank's loan rejection model must explain why an applicant was denied credit.
- *Solution:* Use interpretable models (like Decision Trees) or apply explainability tools like SHAP, LIME, or feature importance plots.

Challenges in Machine Learning (Contd…)

4. Ethical Considerations: ML systems can unintentionally perpetuate or amplify biases present in data or design.

Key concerns:

- Algorithmic bias and discrimination
- Privacy violations
- Misuse of predictive models (e.g., surveillance)
- Lack of informed consent

- *Example:* A facial recognition system performing poorly on darker-skinned individuals due to biased training data.
- *Solution:* Diverse datasets, fairness audits, transparent practices, and ethical review processes.